MIL-I-46520D (AR)
25 July 1979
SUPERSEDING
MIL-I-46520C (PA)
18 March 1975

# MILITARY SPECIFICATION

## IGNITER, PRIMER, ELECTRIC, M74
## PARTS FOR

This Specification is approved for use by the US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers the metal parts for one type of igniter designated as Igniter, Primer, Electric, M74.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking of Inert Ammunition Components

FSC: 1390

**Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.**

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DRAWINGS

ARRADCOM

7548483 - Disc, Closing
8831028 - Plug, Closing
8840275 - Body, Igniter
8862384 - Disc, Closing

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI - B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standard Institute (ANSI), 1430 Broadway, New York, New York 10018).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawings (dwgs.) 7548483 (or 8862384), 8831028 and 8840275, as applicable, and with all requirements specified in applicable specifications.

3.3 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

| PART DESCRIPTION | DRAWINGS | QUANTITY |
|---|---|---|
| Disc, Closing | 7548483 or 8862384 | 50 |
| Plug, Closing | 8831028 | 50 |
| Body, Igniter | 8840275 | 50 |

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-(See MIL-A-48078)

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-(See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.



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TABLE I - FIRST ARTICLE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

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| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Parts For Igniter, Primer, Electric, M74 | | | | See Below |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Plug, Closing (Dwg. 8831028) Examination for defects | 50 | | 3.2 | 4.4.2.2 |
| | Body (Dwg. 8840275) Examination for defects | 50 | | 3.2 | 4.4.2.3 |
| | Disc, Closing (Dwg. 7548483 or 8862384) Examination for defects | 50 | | 3.2 | 4.4.2.4 |

NOTES:

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4.4.2.1 Visual inspections.-In the event of a dispute, four power (4X) magnification shall be used in performing visual inspections for such defects as burrs and foreign matter. The roughness comparison specimens prescribed by ANSI - B46.1 shall be used as a basis for surface roughness determination.



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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46520D (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.2 | Plug, Closing | | | | 8831028 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 8840277 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Pitch diameter of thread | | 0.40% | 3.2 | Gage |
| 102 | Hole obstructed or missing | | 0.40% | 3.2 | Visual |
| Minor | | | | | |
| 201 | Major diameter of thread | | 0.65% | 3.2 | Gage |
| 202 | Hole diameter, min. | | 0.65% | 3.2 | Gage |
| 203 | Diameter of countersink | | 0.65% | 3.2 | Gage |
| 204 | Total length, max. | | 0.65% | 3.2 | Gage |
| 205 | Slot missing | | 0.65% | 3.2 | Visual |
| 206 | Slot width | | 0.65% | 3.2 | Gage |
| 207 | Slot depth | | 0.65% | 3.2 | Gage |
| 208 | Chamfers missing or incorrect | | 0.65% | 3.2 | Visual |
| 209 | Evidence of poor workmanship | | 0.65% | 3.4 | Visual |

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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46520D (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.3 | Body, Igniter | | | | 8840275 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 8840277 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Pitch diameter of external thread, min. | | 0.40% | 3.2 | Gage |
| 102 | Major diameter of external thread, min. | | 0.40% | 3.2 | Gage |
| 103 | Pitch diameter of internal thread, max. | | 0.40% | 3.2 | Gage |
| 104 | Minor diameter of internal thread, max. | | 0.40% | 3.2 | Gage |
| 105 | Width of wrench slot, min. | | 0.40% | 3.2 | Gage |
| 106 | Burrs in screwdriver slot | | 0.40% | 3.2 | Visual |
| 107 | Grooves in base missing, damaged, incomplete, or incorrectly formed | | 0.40% | 3.2 | Visual |
| 108 | Surface finish of ignition element cavity improper | | 0.40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Pitch diameter of external thread, max. | | 0.65% | 3.2 | Gage |
| 202 | Major diameter of external thread, max. | | 0.65% | 3.2 | Gage |
| 203 | Pitch diameter of internal thread, min. | | 0.65% | 3.2 | Gage |
| 204 | True position of hole through base with ignition element cavity | | 0.65% | 3.2 | Gage |
| 205 | Length of internal thread | | 0.65% | 3.2 | Gage |
| 206 | Width of external thread undercut, max. | | 0.65% | 3.2 | Gage |
| 207 | Length to external thread | | 0.65% | 3.2 | Gage |
| 208 | Diameter of ignition element cavity | | 0.65% | 3.2 | Gage |

NOTES:

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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46520D (AR)

| PARAGRAPH | TITLE | | SHEET 2 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.3 | CONTINUATION SHEET | | | | 8840275 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 8840277 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| 209 | Diameter of hole through base | | 0.65% | 3.2 | Gage |
| 210 | Diameter of flange | | 0.65% | 3.2 | Gage |
| 211 | Total length | | 0.65% | 3.2 | Gage |
| 212 | Thickness through bottom | | 0.65% | 3.2 | Gage |
| 213 | Diameter of exterior threaded area including thread undercut | | 0.65% | 3.2 | Gage |
| 214 | Thickness of flange | | 0.65% | 3.2 | Gage |
| 215 | Depth of wrench slot | | 0.65% | 3.2 | Gage |
| 216 | Outside diameter on front of body | | 0.65% | 3.2 | Gage |
| 217 | Groove pitch | | 0.65% | 3.2 | Gage |
| 218 | Groove depth, max. | | 0.65% | 3.2 | Gage |
| 219 | Presence of burrs | | 0.65% | 3.4 | Visual/Manual |
| 220 | Chamfers missing or incorrect | | 0.65% | 3.2 | Visual |
| 221 | Radii missing or incorrect | | 0.65% | 3.2 | Visual |
| 222 | Surface finish improper | | 0.65% | 3.2 | Gage |
| 223 | Evidence of poor workmanship | | 0.65% | 3.4 | Visual |

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# CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46520D (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.4 | Disc, Closing | | | | 7548483 or 8862384 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 8840277 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | None defined | | | | |
| Minor | | | | | |
| 201 | Diameter, min. | | 0.65% | 3.2 | Gage |
| 202 | Thickness (See Note 1) | | 0.65% | 3.2 | Gage |
| 203 | Evidence of poor workmanship | | 0.65% | 3.4 | Visual |

NOTES: 1. Thickness measurement applies only to Closing Disc, dwg. 7548483.

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# CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46520D (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.5 | Cartons, Packed, Prior to Sealing | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | None defined | | | | |
| Minor | | | | | |
| 201 | Any support missing or damaged | | 0.65% | 3.1 | Visual |
| 202 | Separator(s) missing or damaged | | 0.65% | 3.1 | Visual |
| 203 | Improperly packed | | 0.65% | 3.1 | Visual |

NOTES:

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# CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46520D (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.6 | Cartons, Packed, After Sealing | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | None defined | | | | |
| Minor | | | | | |
| 201 | Contents loose | | 0.65% | 3.1 | Manual |
| 202 | Carton damaged, cut or gouged through all layers | | 0.65% | 3.1 | Visual |
| 203 | Carton not properly sealed | | 0.65% | 3.1 | Visual |
| 204 | Marking missing, incorrect or illegible | | 0.65% | 3.1 | Visual |

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

## 5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-The closing plug, body and closing disc shall be packaged and packed, Level B, in accordance with the requirements of MIL-STD-1169.

5.2 Marking.-Marking shall be in accordance with the requirements of MIL-STD-1169.

## 6. NOTES

6.1 Ordering data.-See MIL-A-48078.

6.2 Submission of designs for approval.-Submit equipment designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801 (see 6.2.3 of MIL-A-48078).

6.3 Delivery of data cards to the technical activity.-The following address is to be added to those listed in the contract (Form DD 1423) for delivery of one copy of each Ammunition Data Card prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E, Dover, New Jersey 07801.

6.4 Changes from previous issue.-Due to the extensiveness of the changes, asterisks are not used in this revision to identify changes with respect to the previous issue.

Custodian:
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Preparing Activity:
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Project Number: 1390-A215



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